Packing Process

Last updated: 02/05/2026

- Cubed vs Non-cubed packing
- System vs User directed packing
- Cross Reference oLPNs
- Routing
- Packing Flow Diagram
- Pack Station Flow Diagrams
  - Pack oLPN
  - Put to Store - Cart Flow
- Configuration Summary
  - Pack Strategy
  - Pack Criteria
  - Pack Override Criteria
  - Pick Pack Parameters
- Accept Input/Display Information
  - Display oLPN Type and Size
  - Accept Printer Location
  - Accept Picked Container
  - Accept Tote
  - Accept iLPN
  - Accept oLPN
  - Accept Packing Location
  - Accept Pack Cart
  - Accept Pick Label
  - Accept Slot Id
  - Pack Verification
  - Accept oLPN
  - Accept Item
  - Verify Item
  - Enter Regulation Set
  - Accept Quantity
  - Accept Unique (Serial Numbers) or Non-Unique Attributes (Catch Weight)
  - Hazardous Material Packing Data Capture

Copyright © 2026 Manhattan Associates, Inc.

- Accept Asset Serial Number
- Verify/Accept oLPN
- Pack Instructions
- Make Pack Cart
  - Cart ID
  - Container ID
  - Slot ID
- Exceptions
- Weighing
- Routing
- Re-routing
- Sortation Into Transit Tote (Put to Zone)
  - Accept Input/Display Information
- Unit Sorter Dynamic Packing
- Packing Location Assignment Screen
- Packing Location Dedication Screen
- Packing Location Capacity Screen
- Asynchronous Updates
- Asynchronous Pack Station UI

Packing is the process of gathering and packaging items to prepare them for shipment to the customer. It involves choosing the right item(s) in the right quantities and packaging them into a shipping container, namely an oLPN. The packing functionality is designed to allow clients to achieve 100% shipping accuracy while increasing picking efficiency.
WM offers a single packing service for all different flavors of packing transactions, such as Put to Store, Pack Cubed/Non-Cubed from Tote/iLPN and Bulk Packing for Retail/Ecom. Both User initiated and Task initiated pack processes are supported.
The packing process can be initiated by scanning any of the drivers below, where the subsequent flow is determined by the packing criteria definition.

- iLPN
- Tote
- oLPN
- Packing Location
- Pack Cart
- Pick Label Id

The packing process can be done either:

- As a single step, where the items are picked from the location and directly packed into the oLPN.
- As a 2-step process, where the items are first picked from the location into a tote/iLPN and then later packed into the oLPN from

Copyright © 2026 Manhattan Associates, Inc.

the picked tote or items are first picked from the location into a tote/iLPN and transferred to inventory to the packing location and then later packed into the oLPN.

The various packing options and their supporting features during the packing process are further explained below in detail.

Cubed vs Non-cubed packing

Cubed Packing

In this flow, the desired oLPN and the contents to be packed as part of the oLPN are decided by the cubing process for the allocated inventory. The user verifies the item and quantity into the right oLPN during the packing process.

Non-Cubed Packing

In this flow, the system just allocates the inventory, but the user creates the oLPN and decides the contents by verifying the item and quantity to be packed. The oLPN number can either be system generated or manually provided by the user based on the packing criteria configuration.

System vs User directed packing

System directed packing

In this packing flow, the user is automatically directed by the system as to what contents needs to be packed. This is supported only in the put to store flow where the inventory that is picked into the tote is satisfying multiple stores for which there are separate store pack locations. The system would direct the user by the store pack location sequence whereby the user completes all the packing for a location. Additionally, if the "Do you want the system to display/read out the next pack location?" Pack Criteria parameter is enabled, the system informs the user of the next pack location in the sequence while the current pack task detail is being executed.

User directed packing

In this packing flow, the user starts the packing by scanning the item from the picked container upon which the system finds the details for the tote/item combination. Once the details are fetched, the subsequent verification of the location, quantity and oLPN are based on what is being enabled in the packing criteria.

Cross Reference oLPNs

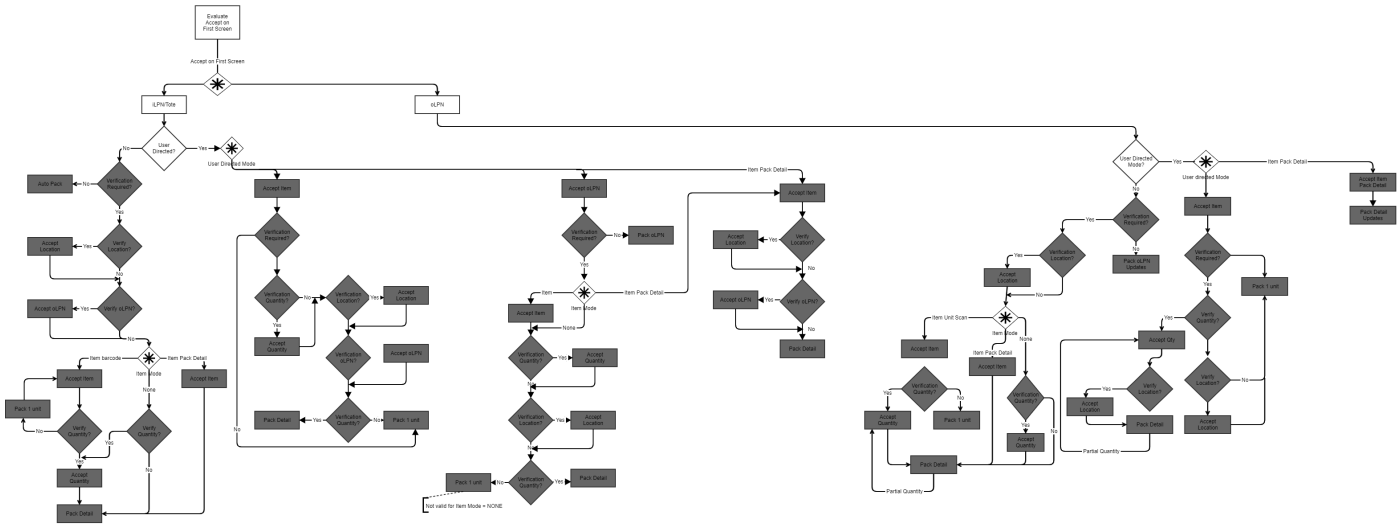
The cross reference oLPNs are nothing but an alternate way of identifying the actual or physical oLPN that will ultimately get packed and shipped. These are typically used in scenarios where either the containers have a permanent physical label attached to them that serves as a cross reference to the actual oLPN or when operations delay the printing of the actual label till the final packing is complete. The cross-reference labels get associated with the actual oLPN when they get scanned for the first time during the packing process. The ability to use the cross reference is controlled by a parameter in the packing criteria configuration.

Routing

Once the oLPN is packed, the system provides the ability to perform the routing and determine the shipment information like the ship Via, tracking number, route, rates and labels. This process involves calling the routing logic to get the ship via, and the rate shop group followed by calling the parcel to get the tracking number, rates and labels.

Packing Flow Diagram

Copyright © 2026 Manhattan Associates, Inc.



Pack Station

Graphical packing stations are widely used in the e-commerce world and are proven to increase packing accuracy, speed and reduce fatigue for the packers in a long run. Manhattan Active WM provides a graphical state-of-the-art, touch-screen UI that supports packing single oLPN and also packing oLPNs from totes.

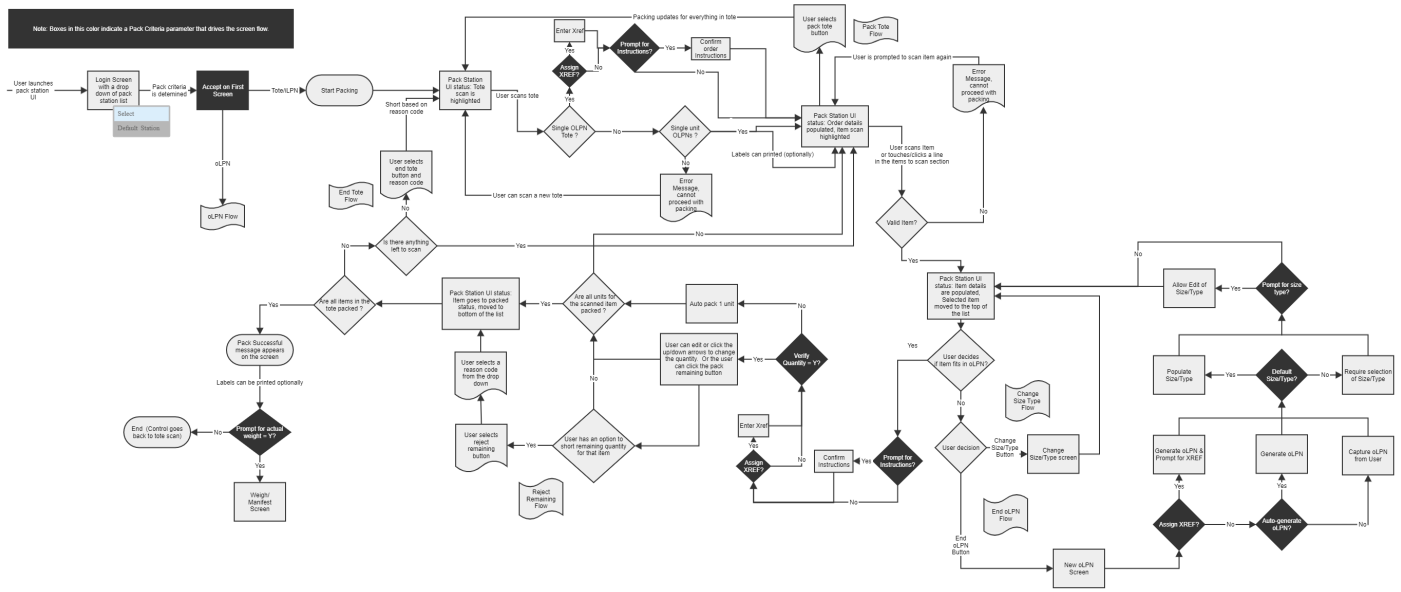
See the corresponding resource IDs for the following buttons available on the Pack Station screen:

- **Pack Remaining**: sceui::pickpack::packStation::packremaining
- **Reject Remaining**: sceui::pickpack::packStation::rejectremaining
- **Pack Tote**: sceui::pickpack::packStation::packtote
- **End Tote**: sceui::pickpack::packStation::endtote
- **End oLPN**: sceui::pickpack::packStation::endolpn
- **Change Type/Size**: sceui::pickpack::packStation::changeoLPNtype
- **Reset**: sceui::pickpack::packStation::resetpacking
- **Reprint**: sceui::pickpack::packStation::reprint

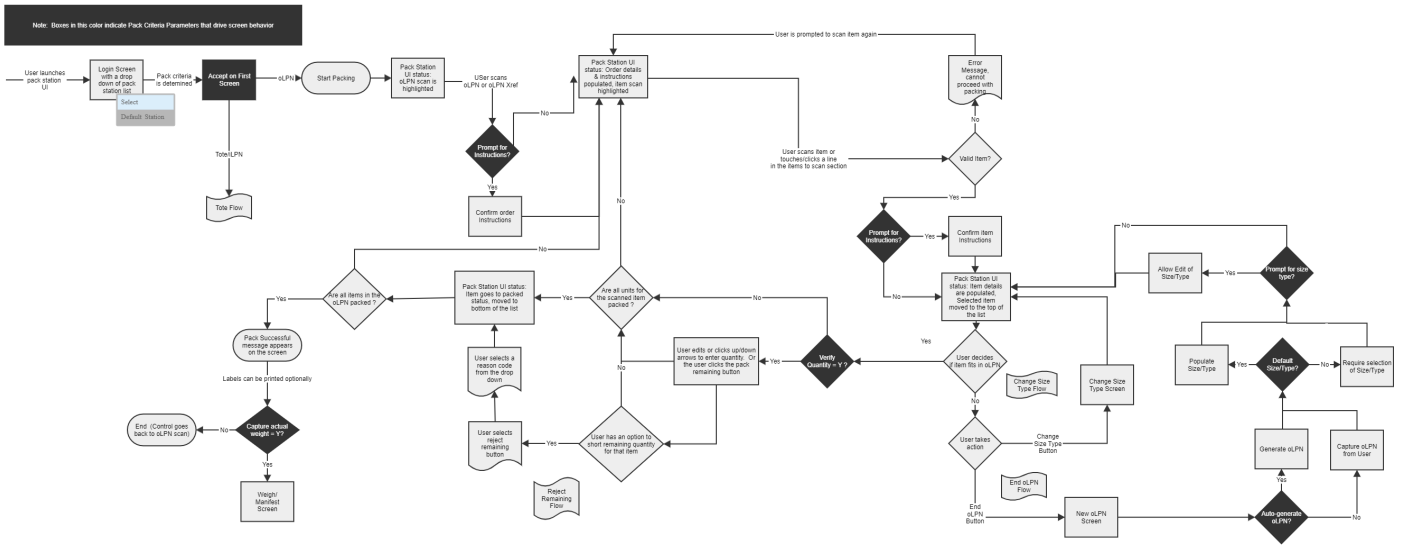
Pack Station Flow Diagrams

Pack oLPNs from a Tote

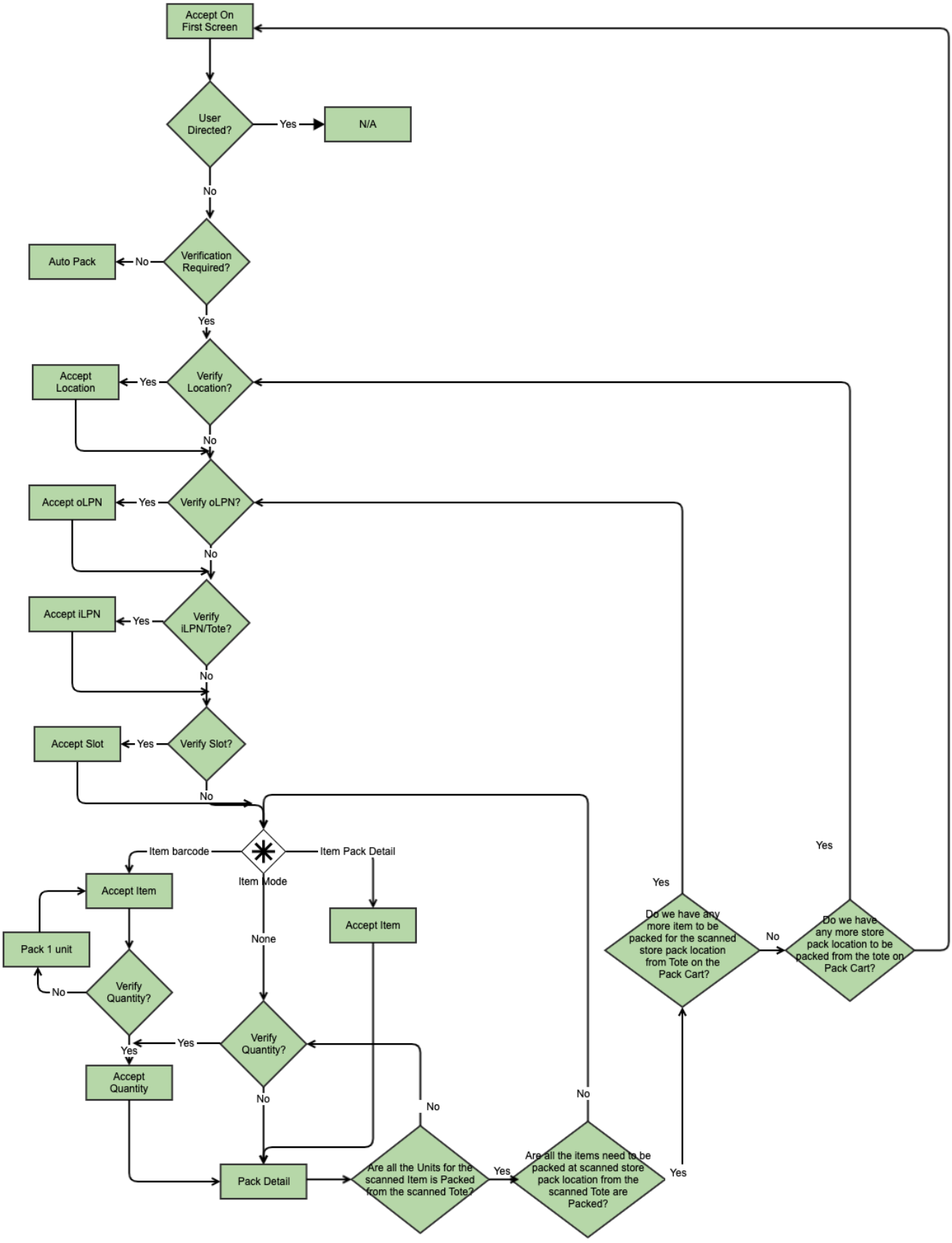
Copyright © 2026 Manhattan Associates, Inc.



Pack oLPN



Put to Store - Cart Flow


Accept On First Screen
User Directed?
Yes
N/A
No
Verification Required?
No
Auto Pack
Yes
Verify Location?
Yes
Accept Location
No
Verify oLPN?
Yes
Accept oLPN
No
Verify iLPN/Tote?
Yes
Accept iLPN
No
Verify Slot?
Yes
Accept Slot
No
Item barcode
Item Pack Detail
Item Mode
Accept Item
Accept Item
None
Pack 1 unit
Verify Quantity?
No
Yes
Verify Quantity?
Yes
Accept Quantity
No
Pack Detail
Are all the Units for the scanned Item is Packed from the scanned Tote?
No
Yes
Are all the items need to be packed at scanned store pack location from the scanned Tote are Packed?
No
Yes
Do we have any more item to be packed for the scanned store pack location from Tote on the Pack Cart?
Yes
No
Do we have any more store pack location to be packed from the tote on Pack Cart?
Yes

Configuration Summary

Pack Strategy

The pack strategy defines the business configuration for the packing process. This defines the parameters which need to be configured at the strategy level, and links one or more pack criteria depending on the rule defined. More information can be found

Pack Criteria

Pack criteria define the packing process through a set of parameters. The set of parameters determines the screen flow, the validation and post process. Pick Criteria is tied to the strategy via rules. More information can be found

Pack Override Criteria

These are parameters that can be overridden for the criteria chosen based on the rule. More information can be found

Pick Pack Parameters

This includes parameters that drive functionality at component level.

Accept Input/Display Information

Below are the various screens that come across during the packing process. The sequence of display of these screens depends on the packing criteria configuration.

Display oLPN Type and Size

This screen displays when the parameter 'Display oLPN Type/Size' = 'Y'. This shows the information about the oLPN description, number of oLPNs, and their corresponding type and size at the start of the packing option. This helps users to have the right size and sufficient oLPN boxes needed for the packing process.

Display information

- Displays the task id during tasking mode
- Displays the count of unique oLPNs for the task if in tasking mode or for the scanned tote from task details
- Displays the container type and container size of the oLPN. This will be displayed for every change of oLPN and when the first detail of that oLPN is being packed.

Actions

This screen supports the following actions

- Refresh - Clears the container entered by the use
- Exit - Exit to the main menu

Accept Printer Location

The system accepts the printer location at the beginning of packing, and uses the respective printer configured in this location for printing labels or documents configured on the pack print criteria/pack override print criteria.

Accept Picked Container

Based on the value configured in packing criteria "Accept on first screen", systems determine what to accept on the first screen of packing. This will be the driver for the packing process.

Copyright © 2026 Manhattan Associates, Inc.

Accept Tote

The system accepts the tote information as a driver when the packing criteria "Accept on first screen" is configured as "Tote".

Display information

Displays the task id when invoked from tasking.

Tote Validations

Error Code	Error Message	Message Type	Error Level	Validations
PPK:0010	Tote id doesn't exist	System	Error	When the user tries to scan a tote that doesn't exist, the inventory component
PPK:0002	Invalid Tote Status	System	Error	When the user tries to scan a tote, it is not picked completely.
PPK:X001		System	Error	When the user tries to scan a tote, it doesn't have picked inventory.
PPK:X002	No Eligible oLPNs to be Packed	System	Error	When a user tries to scan a tote, non-eligible oLPNs are packed out of the tote

Actions

This screen supports the following actions -

- Refresh - Clears the container entered by the use
- Exit - Exit to the main menu

Accept iLPN

The system accepts the iLPN information as a driver when the packing criteria "Accept on first screen" is configured as "iLPN".

Display information

Displays the task id when invoked from tasking.

iLPN Validations

Error Code	Error Message	Message Type	Error Level	Validations
PPK:	iLPN does not exist	System	Error	When the user tries to scan an iLPN, it doesn't exist in the Inventory component.
PPK:	Invalid iLPN status	System	Error	When the user tries to scan an iLPN, it is not picked completely.
PPK:		System	Error	When a user tries to scan an iLPN, which doesn't have picked inventory.
PPK:	No Eligible oLPNs to be Packed	System	Error	When a user tries to scan an iLPN and no eligible oLPNs are to be packed out from iLPN

Actions

This screen supports the following actions:

- Refresh - Clears the container entered by the use
- Exit - Exit to the main menu

Accept oLPN

The system accepts the oLPN information as a driver when the packing criteria "Accept on first screen" is configured as "oLPN". This is applicable in a flow where the inventory is picked into an oLPN directly but the oLPN is in Picked status. The picked oLPN is further taken to a packing station from where the packing happens.

Display Information

Displays the task id when invoked from tasking.

oLPN Validations

Error Code	Error Message	Message Type	Error Level	Validations
PPK:	oLPN does not exist	System	Error	When scanned, oLPN doesn't exist at LPN component.
PPK:	Invalid oLPN status	System	Error	When scanned, oLPN is not in picked status

Actions

This screen supports the following actions:

- Refresh - Clears the container entered by the use

Copyright © 2026 Manhattan Associates, Inc.

- Exit - Exit to the main menu

Accept Packing Location

The system accepts the Packing Location information as a driver when the packing criteria "Accept on first screen" is configured as "Pack Location".

Display Information

Displays the task id when invoked from tasking.

Packing Location Validations

Error Code	Error Message	Message Type	Error Level	Validations
PPK:	Invalid Packing Location	System	Error	When a user tries to scan a packing location of type other than "MHE_PUTWALL" or "MHE_UNIT_SORTER" or "OUTBOUND_MANUAL_SORTER" or "PACK_STATION" or "OTHER_PACK_LOCATION"
PPK:	No allocations for Pack Location	System	Error	When no open pack task details associated with the scanned location

Actions

This screen supports the following actions:

- Refresh - Clears the packing location entered/scanned by the user
- Exit - Exit to the main menu
- Close Location - On taking this action button, the system accepts a packing location and clears the dedications when no more open packing from the scanned/entered location.

Accept Pack Cart

The system accepts the Pack Cart information as a driver when the packing criteria "Accept on first screen" is configured as a "Pack Cart".

Packing Location Validations

Error Code	Error Message	Message Type	Error Level	Validations
PPK:	Invalid Pack Cart Id	System	Error	When the Cart Id not found on the Pack Task
PPK:	Container/Cart <Container ID/Cart ID> is being worked upon by another user <USER_ID>	System	Error	When a user scans a Cart ID which is already being worked upon by another user. Task is In-Progress and is assigned to another user

Accept Pick Label

The system accepts the Pack Cart information as a driver when the packing criteria "Accept on first screen" is configured as "Pick Label".

Packing Location Validations

Error Code	Error Message	Message Type	Error Level	Validations
PPK:0926	No Allocations for Pick Label	System	Error	When no open pack task details associated to the scanned Pick Label Id

Accept Slot Id

This screen displays when "Verify Slot?" is set as true and the Pack Task details. Slot Id is not Null

Display information

This screen displays the following information:

- Pack Cart Id
- oLPN Id
- Picked Container Id (Tote/iLPN)

Slot Validations

Error Code	Error Message	Message Type	Error Level	Validations
PPK:	Slot <Slot ID> does not match	System	Error	When the scanned slot id is different than the suggested slot id

Copyright © 2026 Manhattan Associates, Inc.

Pack Verification

The system allows you to verify the key attributes like location, item, quantity and oLPN during the packing process to ensure accuracy, when packing criteria 'Verification Required' = 'Y'. The need to verify these attributes is configurable in packing criteria by various 'Verification Modes'.
The packing criteria 'User Directed' parameter defines who controls the sequence of packing of items into an oLPN, either it's user or system.

User Directed Flow

When the packing criteria 'User Directed' parameter is set as 'Y', the user controls the sequence of packing of items into an oLPN. User initiates packing by scanning either an oLPN or Item.

Accept oLPN

This screen displays, when packing criteria 'User Directed Mode' = 'oLPN'. This screen directs the user to start-off packing with an oLPN scan. This is a possible flow if either the user has labels printed at wave time or if the labels are batch printed upon Tote scan. Upon scanning of the right oLPN system, it identifies the items to be packed into an oLPN.

Display Information

This screen displays the following information:

- Scanned Pick Container

oLPN Validations

Error Code	Error Message	Message Type	Error Level	Notes
PPK:	oLPN/Order Does Not Exist!	System	Error	When scanned, oLPN doesn't exist at LPN component.
PPK:	Wrong oLPN	System	Error	When oLPN doesn't have any open items to pack out from scanned tote/iLPN
PPK:	oLPN is not eligible to be packed	System	Error	When oLPN is not eligible to be packed based on the oLPN pipeline

Actions

This screen supports the following actions:

- Refresh - Clears the container entered by the use
- Previous - Go back to Accept Tote screen
- Exit - Exit to the main menu

Accept Item

This screen displays when 'User Directed Mode' = 'Item' or 'Item Pack'. This screen directs the user to start-off packing with an item scan. On successful acceptance of the item system determines the oLPN detail to be packed.

Display Information

This screen displays the following information:

- Scanned Pick Container

Item Validations

Error Code	Error Message	Message Type	Error Level	Notes
PPK:	Invalid Barcode Format	System	Error	When the scanned barcode format doesn't match (Barcode type = Item ID)
PPK:	Invalid Item Barcode	System	Error	When the scanned barcode doesn't match the barcode of the item displayed on the screen
PPK:	Invalid Item	System	Error	When scanned, the barcode item doesn't exist at picked container (tote/iLPN)

Actions

This screen supports the following actions:

Copyright © 2026 Manhattan Associates, Inc.

- Refresh - Clears the container entered by the use
- Previous - Go back to Accept Tote screen
- Exit - Exit to the main menu

System Directed Flow

The packing criteria 'User Directed' parameter are set as 'N', the system controls the sequence of packing of items into an oLPN. Once the user scans, picked container system prompts for the verification of item/quantity/oLPN based on the verification modes configured the packing criteria.

Verify Item

This screen is displayed after successful acceptance of a picked container, when the 'Item Verification Mode' is 'Item' or 'Item Pack'. The system identifies the item to be packed into an oLPN from the scanned picked container.

Display Information

This screen displays the following information:

- Scanned Pick Container.
- oLPN scanned at Accept oLPN screen.
- Item to be packed into an oLPN.

Validations

When the user scans an Item Barcode, the system performs the following validation.

Error Code	Error Message	Message Type	Error Level	Notes
PPK:	Invalid Barcode Format	System	Error	When the scanned barcode format doesn't match (Barcode type = Item ID)
PPK:	Invalid Item Barcode	System	Error	When the scanned barcode doesn't match the barcode of the item displayed on the screen
PPK:	Invalid Item	System	Error	When scanned, the barcode item doesn't exist at picked container (tote/iLPN)

Actions

This screen supports the following actions:

- Refresh - Clears the container entered by the use
- Previous - Go back to Accept Tote screen
- Exit - Exit to the main menu

Updates

On successful acceptance of an item and when packing criteria 'Quantity Verification mode' is set as 'None' and oLPN is already accepted, the system performs packing updates. Refer to the Packing Updates section for more details.

Enter Regulation Set

This screen is displayed after scanning a hazardous material. The user selects the regulation set which the hazardous material oLPN needs to be packed with. If the user does not select any regulation set on this screen, the system uses the regulation set defined in the hazardous material entity.

Display Information

This screen displays the following information:

- Select Regulation Set

Validations

When the user selects a regulation set, the system performs the below validation -

Copyright © 2026 Manhattan Associates, Inc.

Error Code	Error Message	Message Type	Error Level	Notes
PPK::0528	No matching regulation set was found for the item.	System	Error	Error prompts when a user selects a regulation set that does not match with what is configured in hazardous material for that item.

Accept Quantity

This screen is displayed when the 'Quantity Verification mode' is enabled and accepts the quantity to be packed.

Display Information

- Scanned Pick Container.
- Scanned Item and its corresponding item description and image.
- The oLPN number is associated with the current details.
- Need and Packed quantity.
  - Need Quantity - The allocated quantity from allocation/task detail for an oLPN.
  - Packed - The quantity that is packed for the detail.

Quantity Validations

Error Code	Error Message	Message Type	Error Level	Notes
PPK:	Quantity should not be zero	System	Error	When entered, the quantity is zero or no quantity entered
PPK:1026	Order line quantity exceeds Original Order line quantity	System	Error	If a user attempts to pack inventory against a Distribution Order Line with a quantity that exceeds its associated Original Order Line's quantity.

Actions

This screen supports the following actions:

- Refresh - Clears the container entered by the use
- Previous - Go back to Accept Tote screen
- Exit - Exit to the main menu

Updates

On successful acceptance of quantity and an oLPN is already accepted, the system performs packing updates. Refer to the Packing Updates section for more details.

Accept Unique (Serial Numbers) or Non-Unique Attributes (Catch Weight)

A screen is displayed to capture unique attributes, for example, serial numbers or non-unique attributes and catch weight.

Hazardous Material Packing Data Capture

The system will display an additional screen to capture packaging details if the oLPN or package contains hazardous material in it.

Display Information

- Overpack (Y/N)
- All in One (Y/N)
- Q Value (Numeric)
- Salvage Pack (Y/N)
- Salvage Pack Quantity (Numeric)
- Dry Ice (Y/N)
- Dry Ice Weight (Numeric)
- Special Permits (Text)

Copyright © 2026 Manhattan Associates, Inc.

Updates

After the user enters all this information, these values are updated in the oLPN.

Accept Asset Serial Number

The solution checks if the oLPN is associated with a serialized asset item (the asset item is defined on the container type/size) when an oLPN is completely packed. If so, the solution prompts you to capture the asset serial number for the packed oLPN. The prompt is displayed only when the following conditions are met:

- The *Do you want to capture the asset serial number for the oLPN?* parameter is enabled.
- An asset serial number is not captured for the oLPN.
- The oLPN's container size is associated with an asset item that is configured with an Attribute Group that tracks a primary, unique attribute during outbound processes.

The prompt is displayed when the oLPN is completely packed. This can be at any of the following touchpoints:

- When all pack task details are completed for a cubed oLPN.
- When the user initiates the *Close oLPN* action to close a non-cubed oLPN or the *End oLPN* action to end a cubed or non-cubed oLPN.

The solution runs the barcode format and attribute group detail validation per the flexible attribute functionality.

Asset Serial Number Validations:

Error Code	Error Message	Message Type	Error Level	Validations
PPK:	{AttributeName) {AttributeValue} is already associated to oLPN {olpnId}	System	Error	The captured attribute (serial number) does not exist for another (unshipped) oLPN associated with the same asset item (if the Unique Within attribute is set to *Item*) or regardless of the asset item (if the *Unique Within* attribute is set to Enterprise). Do not consider canceled oLPNs.
PPK:	{AttributeName) {AttributeValue} not found in master list	System	Error	The solution validates whether the captured attribute exists in Created, Received, In Inventory, Lost, or Consumed status for the Asset Item associated with the oLPN when the *Validate Against Master?* attribute in the attribute group is set to Validate.
PPK:	No duplicates allowed for the unique attribute {attributeName} with value {attributeValue}	System	Error (maximum message level) Non-display (minimum message level)	The solution validates whether there is an existing Inventory Attribute record with the same attribute name/value but for a different Asset Item for the following settings: • *Validate Against Master?* attribute in the attribute group is set to DO_NOT_VALIDATE or VALIDATE_AND_ALLOW_NEW_RECORDS • Unique Within attribute is set to Enterprise

Verify/Accept oLPN

This screen is displayed when 'oLPN Verification Mode' is 'oLPN' or 'oLPN/xref' . It allows the user to verify the oLPN that is suggested in a cubed flow and accepts an oLPN in a non-cubed flow. On successful scan of the oLPN, the system performs the completion updates.

In the non-cubed flow, the user can either enter the oLPN or the oLPN is auto-generated by the system when the "Auto Generate oLPN" packing criteria is enabled.

> **Note:** This input also allows oLPN cross reference scan. Refer cross reference section for more details.

Display Information

This screen displays the following information:

- Scanned Pick Container.
- Scanned Item and its corresponding item description and image.
- The user the user entered quantity at Accept Quantity screen.
- The system identifies the oLPN number that the user should scan.

oLPN Validations

Error Code	Error Message	Message Type	Error Level	Notes
PPK:	oLPN/Order Does Not Exist!	System	Error	When scanned oLPN doesn't exist at LPN component.
PPK:	Wrong oLPN	System	Error	When oLPN doesn't have any open items to pack out from scanned tote/iLPN
PPK:	oLPN is not eligible to be packed	System	Error	When oLPN not eligible to be packed based on the oLPN pipeline

Actions

This screen supports the following actions:

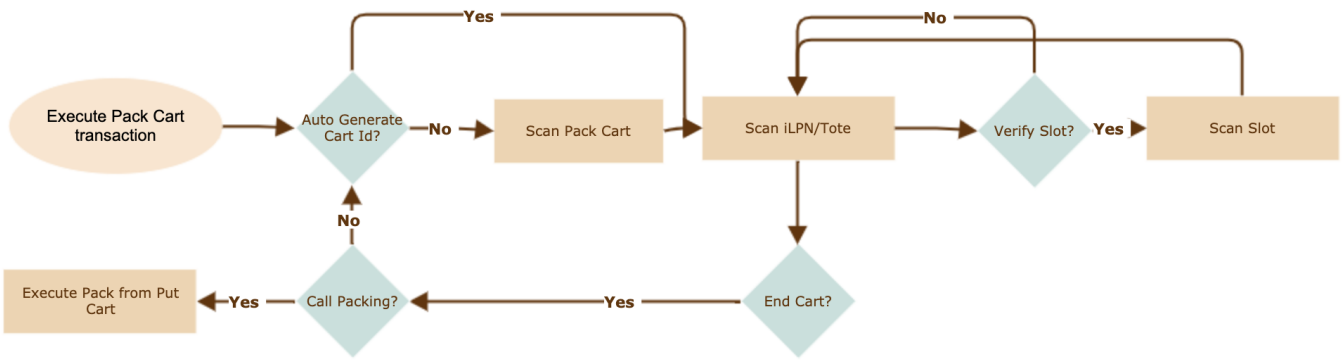
- Refresh - Clears the container entered by the user
- Previous - Go back to Accept Tote screen
- Exit - Exit to the main menu

Pack Instructions

The instructions are prompted at 2 levels. Order instructions are displayed at oLPN scan and order line and item instructions are displayed at item scan.
The instructions are displayed based on the instruction types configured in Pack criteria.

Make Pack Cart

A Pack Cart for Put Store can be built using the Picked Tote or iLPNs.



Following are the various screens that come across during the Make Pack Cart process.

Cart ID

The system accepts the Pack Cart ID.

Cart Id Validations

Error Code	Error Message	Message Type	Error Level	Validations
PPK:	Cart is a Pick Cart	System	Error	When a user scans a pick cart
PPK:	Invalid Cart {id} for Make Pack Cart Transaction	System	Error	When user scans cart id associated to inprogress Pack Task

Actions

This screen supports the following actions:

- Refresh - Clears the cart id entered by the user
- Previous - Exit to the main menu
- Exit - Exit to the main menu

Copyright © 2026 Manhattan Associates, Inc.

Container ID

The system accepts the Picked Tote or iLPN.

Display information

This screen displays the following information:

- Pack Cart Id

Container Id Validations

Error Code	Error Message	Message Type	Error Level	Validations
PPK:	Invalid Container	System	Error	When a user scans an invalid Tote or iLPN
PPK:	Invalid Container Status	System	Error	When the user scans Tote or iLPN, which is not yet Pick Completed
PPK:	PPK:0203 - Container already on Cart	System	Error	When the user scans a Tote/iLPN which is already scanned.

Actions

This screen supports the following actions:

- Refresh - Clears the Container Id entered by the user
- Previous - Exit to the main menu
- Exit - Exit to the main menu
- End Cart - Ends the Making of the Pack Cart
  - If Call Packing is enabled, the system invokes the Packing Transaction for the Pack Cart.

Slot ID

The system accepts the Slot Id.

Display information

This screen displays the following information:

- Pack Cart Id
- Container Id

Container Id Validations

Error Code	Error Message	Message Type	Error Level	Validations
PPK:	Invalid Barcode	System	Error	When a user scans an invalid slot id

Actions

This screen supports the following actions -

- Refresh - Clears the Slot Id entered by the user
- Previous - Exit to the main menu
- Exit - Exit to the main menu

Exceptions

Pack Exceptions (Shortages)

While packing the inventory from a picked container (Tote/iLPN/oLPN), if the user finds less inventory than the system expected, the user can take Pack Exception/Pack Exception Override button to short the packing of the remaining inventory. The system will execute shortage updates based on the chosen Reason Code configuration.

The following options are:

- Short the remaining need
- Create chase at order level to pick into a new oLPN

Copyright © 2026 Manhattan Associates, Inc.

- Create chase at oLPN level to pick into the same oLPN

When the user is doing Pack Exception/Pack Exception Override during packing execution, the reason code has a configuration which may cancel the oLPN (Shortage handling parameter is set as 'Cancel Shortage Requirement' or 'Create inventory need at Order level') and the packing inventory on which exception is happening related to oLPN or oLPNs which belongs to MPS group then the system shows a warning message "oLPN already planned in an MPS group. This operation may cancel the entire MPS group planning. Do you want to continue?" to the user. The user can accept the warning message to continue the execution.

Close Container

While packing the inventory from a picked container (Tote/iLPN/oLPN), when the user finds no inventory in the picked container, the system is expecting a few more items to be sorted from the container. Users can take "Close Container" button to perform shortage updates for all the open pack details. The system will execute shortage updates based on the chosen Reason Code configuration. Available options are as follows:

- Short the remaining need
- Create chase at order level to pick into a new oLPN
- Create chase at oLPN level to pick into the same oLPN

Handling of Exception Inventory

Based on the configuration of "Residual Handling Mode" of the Reason Code chosen for the exception process, the shorted inventory in the picked container can be -

- Consumed
- Move to Lost Inventory bucket
- Keep in the same container without consuming the inventory
  - When the Picked Container is oLPN, the shorted quantity will be transferred to an iLPN. Based on the "iLPN Numbering" configuration in the Reason Code, the system creates an iLPN Id same as oLPN Id or generates a new id as per the next-up configuration.

Weighing

Once the oLPN is packed, the system provides the ability to either auto-weigh the oLPN or capture the weight manually. The below mentioned packing criteria controls the respective behavior

- Auto weigh oLPN - If enabled, copies the estimated weight to actual weight.
- Accept Actual Weight - If enabled, then the system prompts the user to enter the weight.

Routing

Once the oLPN moves to packed status, the system provides the ability to determine the shipment and parcel details as below:

- If the oLPN has an Assigned Ship via which is a Parcel ship via without a tracking number, then the system evaluates the Pick Pack Parameter "Generate tracking number for parcel oLPNs". If enabled, then system verifies tracking number generation strategy is configured and Pick Pack Parameter.
  - If the tracking number generation strategy is not configured, then the system generates the tracking number.
  - If the tracking number generation strategy is configured, the system evaluates the rules and determines the criteria.
    - If the criteria is not selected, then the system generates the tracking number.
    - If the criteria is selected and "Supress Tracking number Generation during Post Pack Routing" is set as 'Yes', then the

system does not generate tracking number.

- If the oLPN does not have an Assigned Ship Via but has a rate shop group ID present, then the rate details are fetched from the parcel component.
- If the oLPN does not have an Assigned Ship Via or rate shop group but the routing strategy ID is populated on the Packing Criteria, then the routing strategy is used to get the Ship Via or rate shop group. Once this is obtained, the tracking number, rates and labels are fetched from the parcel using the Ship via or the rate shop group.

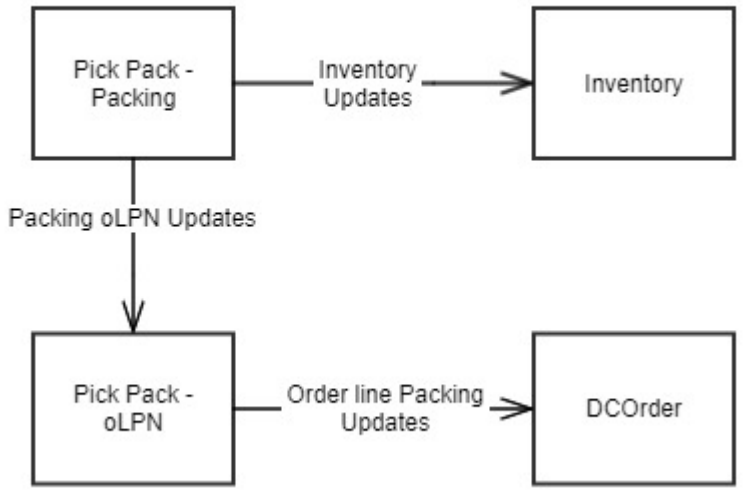
Re-routing

The re-routing for the packed oLPN's happens once the cut off date time of the oLPN is exceeded.

- If oLPN is past the scheduled cut off time (system time more than cut off time calculated Cut off Timestamp) and oLPN has Assigned Ship Via along with a Tracking Number
  - Then continue packing without re-routing
- If oLPN is past the scheduled cut off time (system time more than cut off time calculated Cut off Timestamp) and "Routing Strategy Id" is NULL and there is no Rate Shop Group Id associated with oLPN
  - Then continue with packing without re-routing
- Else the if oLPN is past scheduled cutoff time (system time more than cut off time calculated Cut off Timestamp) and "Routing Strategy Id" is NULL and Rate Shop Group Id is associated with the oLPN
  - Then make a call to Parcel with Rate Shop Group ID to perform the rate shopping
- Else, if oLPN is past the scheduled cut off time (system time more than cut off time calculated Cut off Timestamp) and "Routing Strategy Id" is not NULL then
  - Invoke Routing with corresponding Routing Strategy for the oLPN to determine the new Ship Via or new Rate Shop Group
  - The new Rate Shop Group or Ship Via determined by Routing is used to call Parcel to perform the rate shopping and obtain the tracking number, rate and Labels

Updates

On successful acceptance of oLPN, the system performs packing updates.

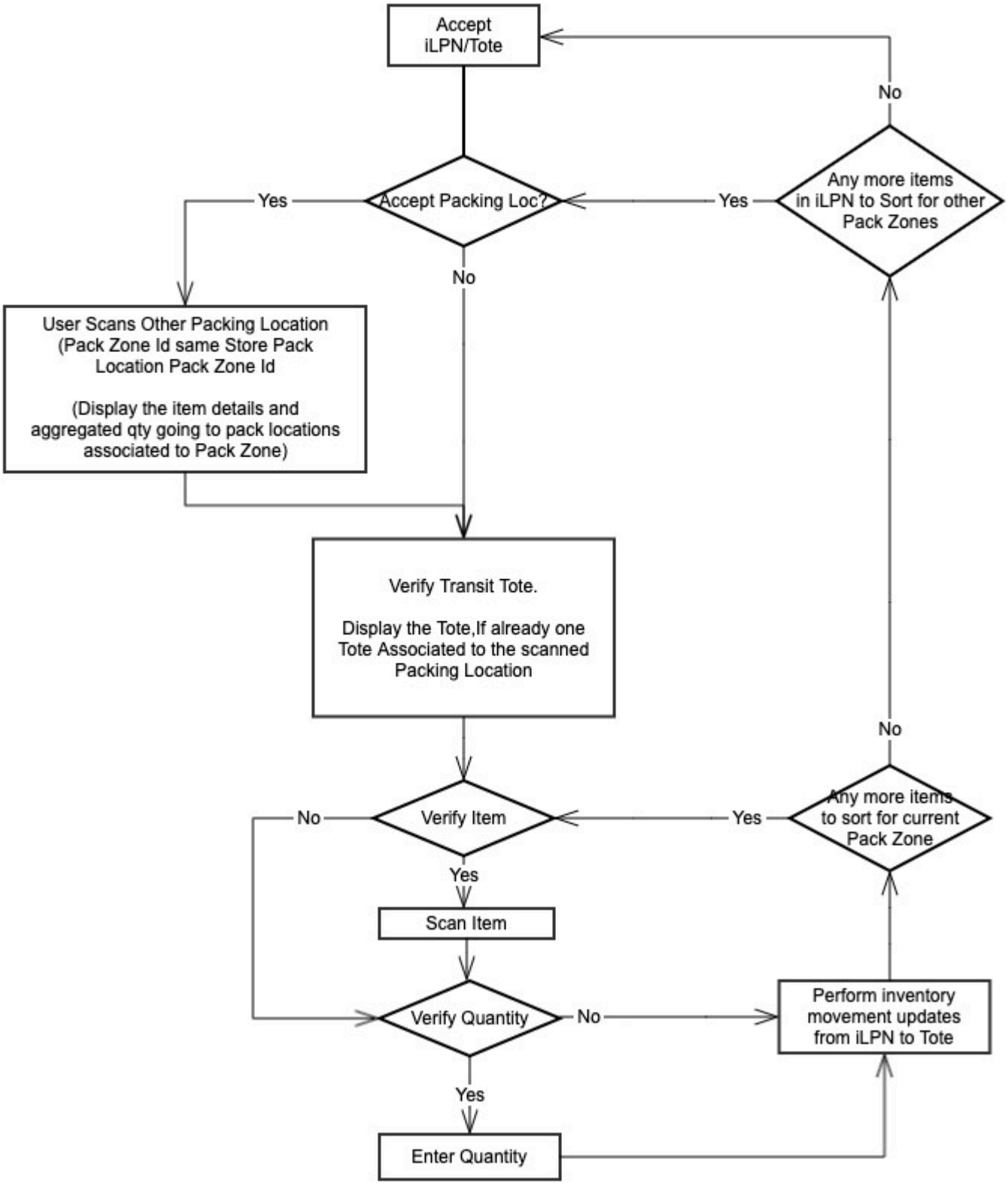


- When an inventory moves from a picked container (Tote/iLPN) to a packing container (oLPN),

- The allocation/task status will be updated(from open to in-progress to complete; depending upon whether the task is in progress or complete).
- The oLPN /oLPN details will be updated as:
  - oLPN details packed-quantity with the entered quantity.
  - oLPN estimated weight and volume in correspondence to the packed quantity.
  - When oLPN is ended or there is no more open allocations or open details for oLPN, then the status of oLPN is "Packed". Otherwise it's in packing.
- Will be reducing the on-hand inventory from the source inventory of tote with the quantity packed. If the entire quantity from a Tote/iLPN is moved, then the status of Tote/iLPN will be of Consumed.
- Order line details will be updated as -
  - The order line packed quantity with the quantity that is packed for the oLPN detail matching the order line.
  - Order's estimated weight and volume in correspondence to the packed quantity.
  - When all the orderline order quantity is packed, the order line status will be packed, otherwise it will be In Packing.
- Various PIXs will be generated to notify the host system on inventory and pack container (oLPN) updates.
- The solution applies an Over-Packed oLPN Condition Code to the packed oLPNs containing a quantity that exceeds the associated Original Order Line quantity.

Sortation Into Transit Tote (Put to Zone)

Sortation into Transit Tote (Put to Zone) is the process of distributing the picked inventory into transit totes assigned to specific Pack Zones for eventual distribution to pack locations designated for stores. Multiple totes are able to be 'active' in the same zone at the same time. A Pack Zone is a grouping of pack locations designated for stores in a retail facility, often physically linked.
Process flow of Sortation into Transit Tote


Accept iLPN/Tote
No
Any more items in iLPN to Sort for other Pack Zones
Yes
Accept Packing Loc?
Yes
No
User Scans Other Packing Location (Pack Zone Id same Store Pack Location Pack Zone Id
(Display the item details and aggregated qty going to pack locations associated to Pack Zone)
Verify Transit Tote.
Display the Tote,If already one Tote Associated to the scanned Packing Location
No
Any more items to sort for current Pack Zone
Yes
No
Verify Item
Yes
Scan Item
Verify Quantity
No
Perform inventory movement updates from iLPN to Tote
Yes
Enter Quantity

To perform the Sortation into a Transit Tote (Put To Zone), set the Pack Criteria parameter "Are you performing Sortation into a Tote?" as Yes. When this parameter has been set as Yes, the system hides some of the Pack Criteria parameters and allows to configure only the parameters applicable to Perfrom Sortation into Transit Tote.
Sortation into Transit Tote is the process that helps the operations to sort the picked into transit tote available pack zone. On scanning the picked container (iLPN/Tote), the system aggregates the need across all the store pack locations per pack zone and directs the user to sort the inventory into transit tote.
When the pack criteria parameter "Do you want to accept location for Transit Tote?" is set as Yes, then the system allows user to scan the Packing Locations of type "Other Pack Locations" and having the Pack Zone Id is the same as the Pack Zone Id for which the system is directed to sort the inventory.

Accept Input/Display Information

Accept iLPN/Tote

On initiating the Tote Sortation process, the system prompts you to accept the picked container (iLPN/Tote)

Validations

Error Code	Error Message	Message Type	Error Level	Validations
PPK::0933	No Open Allocation for Sorting at Pack Zone	System	Error	When there are no Open Task details, having PackingLocationId of PackLocationTypeId = "STORE-PACK" and PackZoneID is not Null

Actions

This screen supports the following actions:

- Refresh - Clears the container entered by the use
- Exit - Exit to the main menu
- Close Transit Tote - To Close the Transit Tote used for Tote Sortation.
- Accept Zone - Allows users to scan the Pack Zone ID, for which the user wants to start the sortation process.

Accept Zone

When the user takes the "Accept Zone" action button on "iLPN/Tote" scan screen, they display a screen to accept the Pack Zone.

Validations

Error Message	Message Type	Error Level	Validation
The field cannot be blank	System	Error	Validate if a value is entered
Invalid Pack Zone	System	Error	Validate it's valid pack zone id

Actions

This screen supports the following actions -

- Refresh - Clears the container entered by the use
- Exit - Exit to the main menu
- Return to Tote Sortation - Return to the Tote/iLPN scan screen to initiate the sortation.

Accept Location

The system allows us to scan other packing locations. On successful rotation of the inventory into transit tote, the system locates the transit tote at scanned location.

Display Information

- Tote/iLPN
  - Scanned Picked Container.

Copyright © 2026 Manhattan Associates, Inc.

- Pack Zone Id
  - Pack Zone Id of the PackLocationId of the Task Details which are sorted.
- Item
- Aggregated Quantity for a given Pack Zone Id, Item Id and Item Attributes.

Validations

Error Message	Message Type	Error Level	Validation
Field cannot be blank	System	Error	Validate if a value is entered
Invalid Barcode Format	System	Error	Validate Packing location barcode.
Invalid Other Packing Location	System	Error	Validate if a valid Packing Location of PackLocationTypeId = "OTHER_PACK_LOCATION"
Invalid Pack Zone {Pack Zone ID of the scanned location }	System	Error	Validate if the scanned location Pack Zone Id = Pack Zone Id displayed over the screen.

Unit Sorter Dynamic Packing

Unit Sorters are a common form of MHE which performs discrete packing of inducted inventory to oLPNs. The inducted inventory comes from tote picks or bulk LPN pulls. Historically, WM dictates to the WCS which inducted inventory must be confirmed to which oLPN. However, certain WCS systems are infused with a level of intelligence to accumulate inducted inventory over time and confirm packing dynamically based on which oLPN/Order can be completed rather than which totes have been inducted. To support this, WM defers the allocation of the bulk picked inventory until packing. As part of the bulk allocation process, the oLPN and Order information are communicated to the WCS. During Pick completion, WM sends the inventory being inducted to the sorters. Finally, WCS controls the decision as to which oLPN to Pack.

To distribute the allocated/picked inventory during packing, at Allocation Strategy, configure "When should the system distribute bulk allocations to the group?" as "Packing".

Once WCS moves the inventory into oLPN, WCS communicates the packing info via PACKING MHE message. This PACKING MHE message should have the Source Container and oLPN/Order Id information along with the item and quantity information.

Packing Location Assignment Screen

This screen displays the packing location assigned to the container after the wave is run. The values related to Assignment Status, Location, Order Planning Run ID, and so on provide more information about the container assignment.

You can navigate to the oLPNs and Location Master screens using hyperlinks. Using Related Links, you can navigate to the Packing Location Capacity Tracking and Packing Location Dedication screens.

Packing Location Dedication Screen

This screen displays the packing location assigned to a container for a specific dedication. Using Related Links, you can navigate to the Packing Location Capacity and Packing Location Assignment screens.

Packing Location Capacity Screen

This screen displays the current location capacity using metrics such as current weight, volume, and quantity. The record is displayed only when the **Track Capacity** parameter is enabled from the Location Master screen. Using Related Links, you can navigate to the Packing location dedication and Packing location assignment screens.

Asynchronous Updates

To experience better performance during the packing process, asynchronous packing updates can be enabled so that the users can continue packing while the majority of updates are made asynchronously.

Note: To enable asynchronous packing updates, set AllowAsyncUpdates = true on the Pack Criteria by using the Update

Pack Criteria API. For more details on this API, refer to the Developer Hub Documentation.

The table below lists supported and unsupported functionalities when asynchronous packing updates are enabled (unsupported functionalities are not limited to those in the below list).

Functionalities supported with asynchronous updates	Functionalities not supported with asynchronous updates
Pack from Tote/iLPN • Mobile cubed packing • Mobile non-cubed packing (Ex: Put-to-Store) • Pack Station UI	Pack picked oLPN
Pack Exceptions • Pack Exception / Pack Exception Override • End oLPN • Close oLPN • Close Container	Pack Pick Label
Unique attribute capture	Pack from Packing Location
Non-unique attribute capture	Pack from Cart
Capture of actual weight	Sortation into Transit Tote (Put-to-Zone)
Chaining to Outbound Putaway	Assigned Instructions
Printing	Chaining to VAS
Routing	Display of the PPK::0252 “Action may lead to Cancellation of Unpicked/Unpacked Inventory.“ message when the user invokes short
Evaluation of Pack Override Criteria	

Asynchronous Pack Station UI

To experience better performance during the Pack Station UI packing process, the AllowAsyncPackStation parameter on the Pack Criteria can be enabled. When enabled:

- Item barcode scan performance is improved because the screen performs validations itself using item data obtained when the picked container is scanned on the screen.
- The screen submits updates asynchronously so the user can continue packing additional inventory while updates are being made.

Note: To enable the above performance improvements, set AllowAsyncPackStation = true on the Pack Criteria by using the Update Pack Criteria API. For more details on this API, refer to the Developer Hub Documentation. Setting AllowAsyncUpdates = true is a prerequisite for setting AllowAsyncPackStation = true. If AllowAsyncUpdates = false, then AllowAsyncUpdates will be set to false by the system.

Note: If a customer has extensibility that relies on updates being made synchronously, then the asynchronous Pack Station UI should not be enabled for that customer.

Handling Asynchronous Update Failures in the Pack Station

When an asynchronous update fails due to an exception, the Pack Station screen alerts the user with the following message: **“Unexpected error occurred. Please reset, scan the container again, and retry packing/shorting the inventory.”**

Steps to Follow:

1. Reset Action:
   - The screen prompts the user to invoke the Reset action.
   - This step allows the picked container to be re-scanned.
   - Once re-scanned, the data displayed in the grid is refreshed, ensuring that systematically packed items are accurately reflected.
2. Operational SOP:
   - An operational Standard Operating Procedure (SOP) must be in place for users to compare what is systemically packed with what is physically packed.
   - Any inventory that is physically packed but not systemically packed in the system (due to the asynchronous update failure) must be re-scanned.

Note: If a Pack Task detail is canceled after a picked container is scanned in the Pack Station, an asynchronous update failure will occur. For customers where the cancelation of an order, order line, oLPN, etc., (post-picking) is common, it is advisable not to enable the asynchronous Pack Station screen for those users.

The table below lists supported and unsupported functionalities when the asynchronous Pack Station UI is enabled.

Functionalities supported with asynchronous Pack Station UI	Functionalities not supported with asynchronous Pack Station UI
Scan of non-composite item barcode or composite item barcode	Capture of actual weight (Weigh and Manifest UI pop-up)
Select Item pop-up when scanned barcode is associated to multiple items in the container	
Display and confirmation of instructions	
Capture and validation of item attributes	
Capture and validation of unique attributes	
Auto-pack based on barcode's quantity (quantity verification disabled)	
Scan item and enter quantity (quantity verification enabled)	
Capture of Cross-reference LPN	
oLPN verification	
Label printing **Note:** Labels print as asynchronous updates are processed so there may be a delay **Note:** The "On Start of oLPN Packing" print trigger condition is not supported	
Use of all available actions: • Pack Remaining • Reject Remaining • Pack Tote / Pack Container • End Tote / End Container • End oLPN • Change Type/Size • Reset • Reprint	
Evaluation of Pack Override Criteria	

Copyright © 2026 Manhattan Associates, Inc.